

UBT832: CONCEPTS IN BIOMEDICAL INSTRUMENTATION

L	T	P	Cr
3	0	2	4.0

Course Objective: The course aims to develop an in-depth understanding of human body subsystems and to impart knowledge about cardiovascular and respiratory system measurements. In addition, the course will also provide understanding of measurement of biomedical processes and medical imaging.

Syllabus

Human Body Subsystems: Brief description of neuronal, Muscular, Cardiovascular and respiratory systems; Their electrical, Mechanical and Chemical activities.

Cardiovascular System Measurements: Electrocardiograph, ECG machines, vector cardiography (VCG), ballisto-cardiography (BCG), measurement of blood pressure, blood flow, cardiac output, cardiac rate, plethysmograph, pacemakers, defibrillators, Heart sounds, Phonocardiograph, Echocardiograph.

Respiratory System Measurements: Measurement of gas volume, respiratory transducers and instruments, respiratory therapy equipment, intermittent positive pressure breathing (IPPB) therapy, artificial mechanical ventilation, accessory devices used in respiratory therapy apparatus.

Measurement of Electrical Activity in Neuromuscular System and Brain: Neuron potential, muscle potential, electromyograph, brain potentials, electroencephalograph.

Analytical Instruments: pH measurement, measurement of $p\text{CO}_2$ & $p\text{O}_2$, calorimeter, blood cell counter, automation of chemical tests, oximeters.

Patient Care, Monitoring and Safety Measures: Elements of intensive care monitoring; Basic hospital systems and components Thermography, ultrasound imaging system, Physiological effect of electric currents, Safety measures; Standards, Codes and practices.

Prosthetics and Orthotics: Introduction to artificial kidney, Artificial heart, Heart lung machine, Limb prosthetics and Orthotics elements of audio and visual aids.

Computer Applications and Biotelemetry: Real time computer applications, Data acquisition and processing; Remote data recording and management.

Laboratory Work (if applicable):

Study of various physiological parameters using multichannel recorder, experiments based on stethoscope, sphygmomanometer, pulse oximeter, ECG, EMG. Respiratory parameters

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study using spirometer for lungs capacity, Ultrasonic characterization study of biological samples

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend the physiology of the heart, lung, blood circulation and respiration.
2. Apply different transducers and various sensing and measurement devices of electrical origin in biomedical applications.
3. Comprehend electrical safety in medical equipment's.
4. Explain different medical imaging techniques.

Text Books

1. Carr, J.J. and Brown, J.M., Introduction to Biomedical Equipment Technology, Prentice Hall (2000) 4th ed.
2. Cromwell, L. and Weibell, F.J. and Pfeiffer, E.A., Biomedical Instrumentation and Measurement, Dorling Kingsley (2006) 2nd ed.

Reference Books

1. Geddes, L.A., and Baker, L.E., Principles of Applied Biomedical Instrumentation, Wiley InterScience (1989) 3rd ed.
2. Khandpur, R.S., Handbook of Biomedical Instrumentation, McGraw Hill (2003) 2nd ed.
3. Webster, J.G., Medical Instrumentation Application and Design, John Wiley (2007) 3rd ed.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes/performance)	30

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